Hanna Bae

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Summary

M.S. student in Electrical Engineering at KAIST (Statistical Inference & Information Theory Lab), specializing in 3D reconstruction, visual SLAM, and multi-view geometry. Sole principal investigator on a competitively awarded National Research Foundation of Korea grant for camera-radar 3D object detection. First author on a journal publication and on long-sequence reconstruction work, co-first author on a further submission, and co-author on three manuscripts currently under review. Experienced in taking vision models from research to deployment under device-level latency and memory constraints (TensorRT, ONNX, CoreML, TinyML).

Education

Korea Advanced Institute of Science and Technology — *Daejeon, Korea*
M.S. in School of Electrical Engineering — *Feb 2025 – Present*

- Statistical Inference & Information Theory Laboratory, advised by Prof. Junmo Kim.
- Coursework: 3D Vision; Computer Vision; Mobile Robotics & Autonomous Driving; Computational Image Generation & Editing; Convex Optimization; Machine Learning; Stochastic Processes.

Gwangju Institute of Science and Technology — *Gwangju, Korea*
B.S. in Electrical Engineering and Computer Science — *Feb 2021 – Feb 2025*

- GPA 4.11 / 4.5 · Converted score 96.1 / 100 · 141 credits completed
- Minors: Mathematics and Artificial Intelligence.

University of California, Berkeley — Summer Session (Data 8, STAT 20) — *Jun – Aug 2022*

Research Grants & Funded Projects

Master's Student Research Grant, National Research Foundation of Korea (NRF) — *Sep 2025 – Aug 2026*
Sole Principal Investigator · RS-2025-25398106 — *KRW 12,000,000*

- SpecFusion: Camera-Radar 3D Object Detection with Specialized Modality Fusion. Authored the proposal and won the grant through national competitive review; the program funds a single student PI with no co-investigators, with full responsibility for research design, execution, budget, and reporting.

Excellent Research Program – Core Research (Type B), NRF — *2026 – Present*
Participating Researcher · RS-2026-25470670 · Physically Consistent 4D World Generation Models

- Leading the lab task on feed-forward 4D scene generation with physical-consistency constraints; project ongoing.

Advanced Research Equipment Commercialization (Pre-Planning), COMPA / MSIT — *Apr – Jun 2026*
Participating Researcher, KAIST subcontract · RS-2026-25536057

- AI-Powered Precision Analysis Platform Integrating Multimodal Fluorescence Imaging and Raman Spectroscopy; KAIST delivered the multimodal AI analysis engine.

AI Technology Utilization Improvement Program, IITP / MSIT — *Mar 2026 – Present*
Participating Researcher · RS-2022-II220184 · Adaptive AI under Evolving Ethics Policy

- Joined as a supporting researcher and now lead the lab task; built the culture-domain Korean NER dataset and domain-adaptation results submitted as a representative lab outcome.

Skills

Programming Python (PyTorch, NumPy, pandas, OpenCV), C++
3D Vision 3D reconstruction, visual SLAM, multi-view geometry, pose estimation, dense correspondence, trajectory evaluation, 3D Gaussian Splatting
Deployment TensorRT, ONNX, CoreML, TinyML, model quantization and pruning, latency–memory profiling
Systems Linux, Git, Docker / Conda, LaTeX
Language Korean (native), English — TOEIC Speaking Intermediate High (140), Jul 2026

Publications

Position Matters in the Overlap: Analyzing Chunk Alignment for Long-Sequence 3D Reconstruction
Hanna Bae, Sojeong Song, Gyojin Han, Sungjae Choi, and Junmo Kim.
Manuscript, under review · First author

Data Preprocessing Methods for Improving Korean Table Question Answering
Hanna Bae, Sanghyuk Won, Jaehong Lee, Woohyuk Jung, and Jihwan Woo.
Information Systems Review, 27(2), 71–87, 2025 · First author

Adaptive Dense Risk Prediction for Consistent Feed-Forward 3D Reconstruction
Sojeong Song, **Hanna Bae**, Chang D. Yoo, and Junmo Kim.
Manuscript, under review

VistaGS: Multi-View Geometry Foundation Priors for View-Consistent Few-Shot 3D Gaussian Splatting
Sungjae Choi, Gyojin Han, **Hanna Bae**, and Junmo Kim.
Manuscript, under review

VGGT-Bridge: Beyond Sequential Pose Graphs via Coarse-Stride Skip Edges
Sungjae Choi*, **Hanna Bae***, Sunghyun Baek, and Junmo Kim. (*equal contribution)
Manuscript, under review · Co-first author

Info3R: Information-Adaptive Test-Time Training for 3D Reconstruction
Sunghyun Baek, **Hanna Bae**, Minchan Kwon, and Junmo Kim.
Manuscript, under review

Research & Industry Experience

CJ OliveNetworks — AI Research Lab · AI Research Intern *Seoul, Korea*
Nov 2024

- Built a Korean table QA dataset of **13,200 question-answer pairs** spanning simple, merged, and composite table structures, distilled from 400K AI Hub tables with an LLM generation pipeline and balanced across six domains.
- Showed that the optimal table serialization depends on structure — Numpy for simple, Text for merged, HTML for composite — lifting average EM/F1 from **43.5/58.1 to 46.5/60.8** over the HTML-only baseline (KoBART), with the largest gain on merged tables (+7.9 EM).
- Developed the internship study into a first-author journal publication (Information Systems Review, 2025).

Autonomous Computing Systems Lab (ACSL), GIST · Research Intern *Gwangju, Korea*
Mar 2023 – Jun 2024

- Researched mobile/edge 3D reconstruction, dense correspondence, and detector-free feature matching for SLAM front ends.
- Benchmarked deployment pipelines (TensorRT, ONNX, CoreML, TinyML), profiling accuracy–latency–memory trade-offs on Jetson.

Project Experience

SpecFusion: Camera-Radar 3D Object Detection · NRF Research Grant · Sole Principal Investigator *Daejeon, Korea*
Sep 2025 – Aug 2026

- Proposed and independently led a nationally funded project (KRW 12M), from research design through budget and reporting.
- Designing a two-path fusion architecture: BEV early fusion with radar-assisted depth sampling and structure-aware depth enhancement, combined with late fusion over pseudo-depth and geometry priors.
- Targeting known radar failure modes — spurious points, key-point missing, and non-positional disturbance — evaluated on nuScenes and View-of-Delft.

Fluorescence-Raman Multimodal AI Analysis Engine · Industry-Academic Project · KAIST subcontract *Daejeon, Korea*
Apr – Jun 2026

- Built a three-stage statistical Raman denoising pipeline (cosmic-ray removal, ALS/airPLS baseline correction, Savitzky-Golay smoothing) with reference-free baseline estimation, recovering reference peaks from spectra dominated by substrate fluorescence.
- Validated deep-learning fluorescence denoising with SAM-based automatic ROI extraction, quantified domain-mismatch limits, and derived fine-tuning and data-collection requirements.
- Designed a metadata-conditioned adapter architecture handling varying magnification (x20/x63/x100) and excitation wavelengths (405/488/561 nm).

OpenVINS-MASt3R Hybrid Visual-Inertial SLAM · Team Project (3 members) · HILTI SLAM Challenge *Daejeon, Korea*
2025

- Diagnosed abrupt pose corruption in static states caused by Z-axis accelerometer noise that MSCKF failed to absorb, then built depth and IMU gating modules for anomaly rejection.
- Added a Hampel median filter (window 11, ~0.03 s) over raw IMU signals; the combined pipeline raised the HILTI score on exp02 from **0 to 0.45** and on exp15 from **6.7 to 22.22**.
- Integrated MASt3R-SLAM global correction — keyframe selection (>1.0 m motion or >3.0 s static), ASMK top-3 retrieval with dense-matching verification, Gauss-Newton pose-graph optimization — lifting exp15 from 6.67 to 15.56 before per-sequence tuning.

SegMatching: Semantic Filtering for Detector-Free Feature Matching · B.S. Thesis *Gwangju, Korea*
2024

- Found through qualitative analysis of TopicFM that dense correspondences concentrate on regions unreliable for pose estimation — sky, foliage, and people.
- Proposed a post-matching filter using ADE20K semantic segmentation masks with per-class stability priors, improving AUC@5 and AUC@10 by 0.8%p and 0.4%p on MegaDepth-1500 over TopicFM.
- Documented the trade-offs: AUC@20 fell 0.2%p from the reduced correspondence count, alongside the added segmentation cost and an outdoor-biased design.

Culture-Domain Korean NER & Group-Fairness Dataset · Government-Funded Research Project *Daejeon, Korea*
Apr 2026 – Present

- Constructed a culture-domain Korean NER dataset from scratch, labeled into the KLUE 6-type schema with an LLM pipeline, converted to char-BIO TSV with lossless round-trip validation, and split with document-level leakage prevention.
- Improved entity F1 on the culture-domain test set through domain-adaptive retraining while retaining performance on KLUE.
- Annotated group attributes (gender, domestic/overseas, entity frequency) across the corpus so that per-group performance gaps can be measured, establishing the basis for fairness-gap reduction work.

Teaching Experience

Teaching Assistant, GIST · Differential Equations (Spring 2023, Fall 2024), Computer Programming (Fall 2023), Computer System Theory and Experiments (Spring 2024).